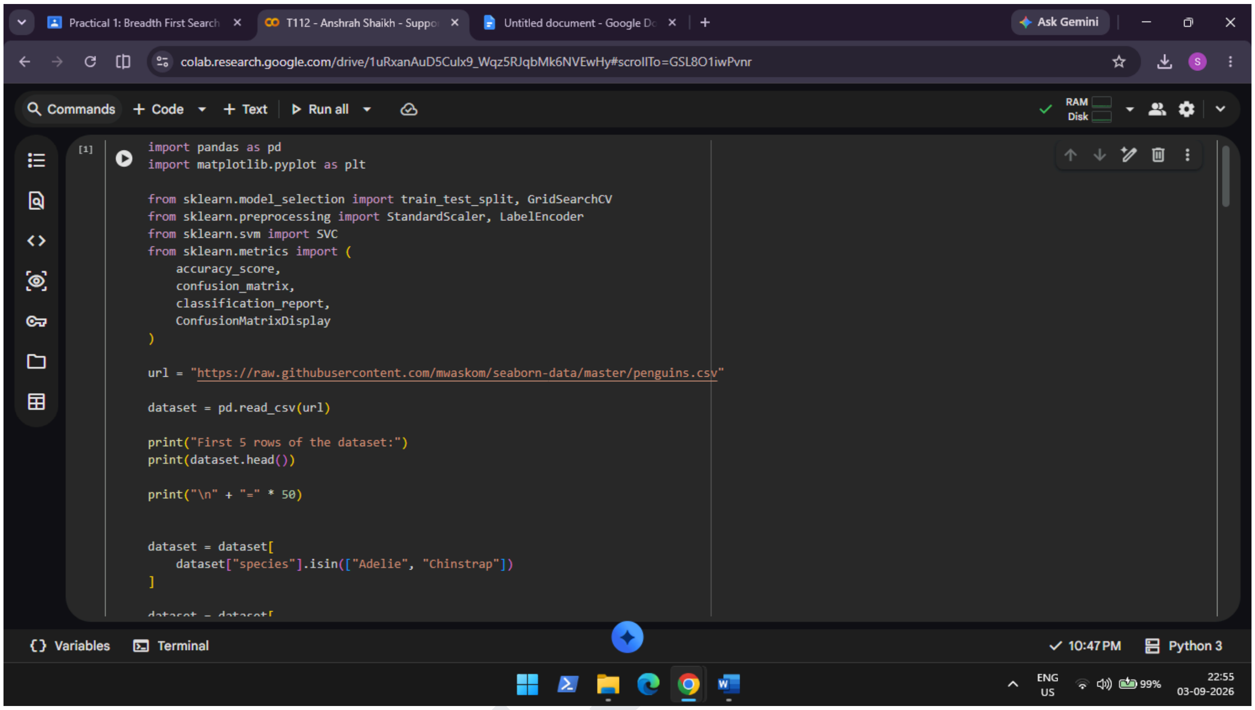
# MVLU COLLEGE

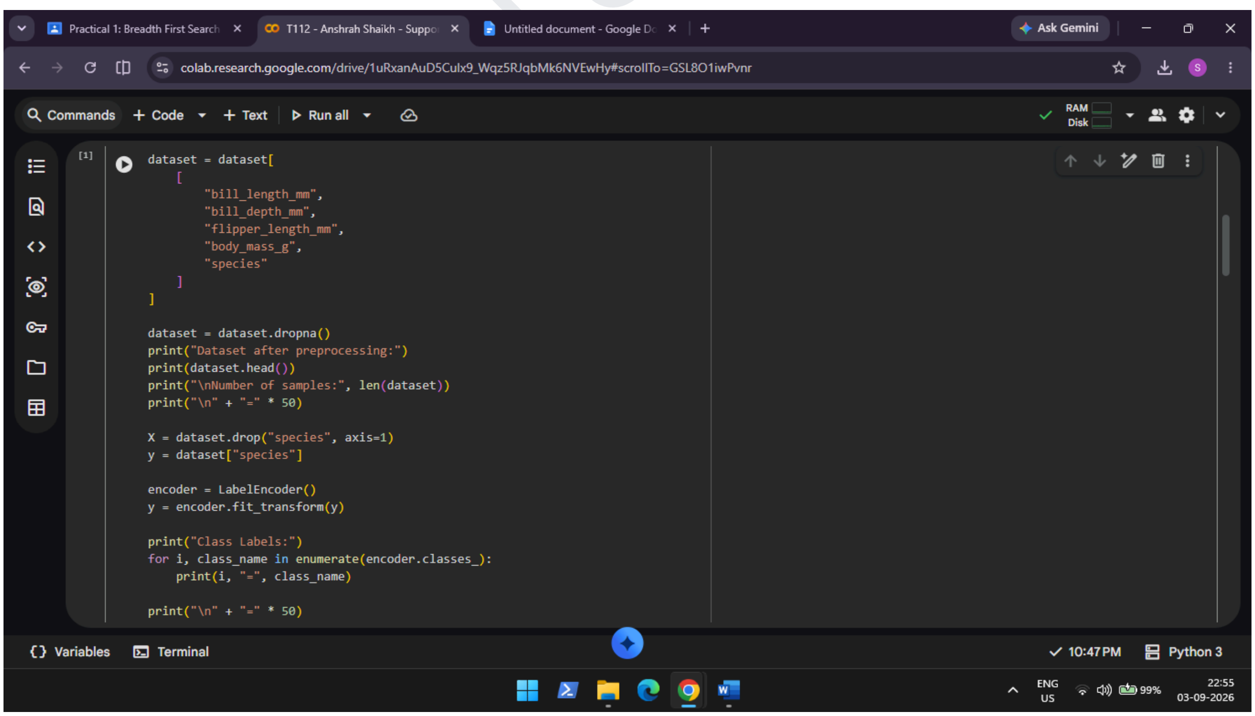
AIM: Support Vector Machines (SVM)

- Implement the SVM algorithm for binary classification.
- Train an SVM model using a given dataset and optimize its parameters.
- Evaluate the performance of the SVM model on test data and analyze the results.

INPUT:



```python
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.svm import SVC
from sklearn.metrics import (
    accuracy_score,
    confusion_matrix,
    classification_report,
    ConfusionMatrixDisplay
)

url = "https://raw.githubusercontent.com/mwaskom/seaborn-data/master/penguins.csv"

dataset = pd.read_csv(url)

print("First 5 rows of the dataset:")
print(dataset.head())

print("\n" + "=" * 50)


dataset = dataset[
    dataset["species"].isin(["Adelie", "Chinstrap"])
]
```



```python
dataset = dataset[
    [
        "bill_length_mm",
        "bill_depth_mm",
        "flipper_length_mm",
        "body_mass_g",
        "species"
    ]
]

dataset = dataset.dropna()
print("Dataset after preprocessing:")
print(dataset.head())
print("\nNumber of samples:", len(dataset))
print("\n" + "=" * 50)

X = dataset.drop("species", axis=1)
y = dataset["species"]

encoder = LabelEncoder()
y = encoder.fit_transform(y)

print("Class Labels:")
for i, class_name in enumerate(encoder.classes_):
    print(i, "=", class_name)

print("\n" + "=" * 50)
```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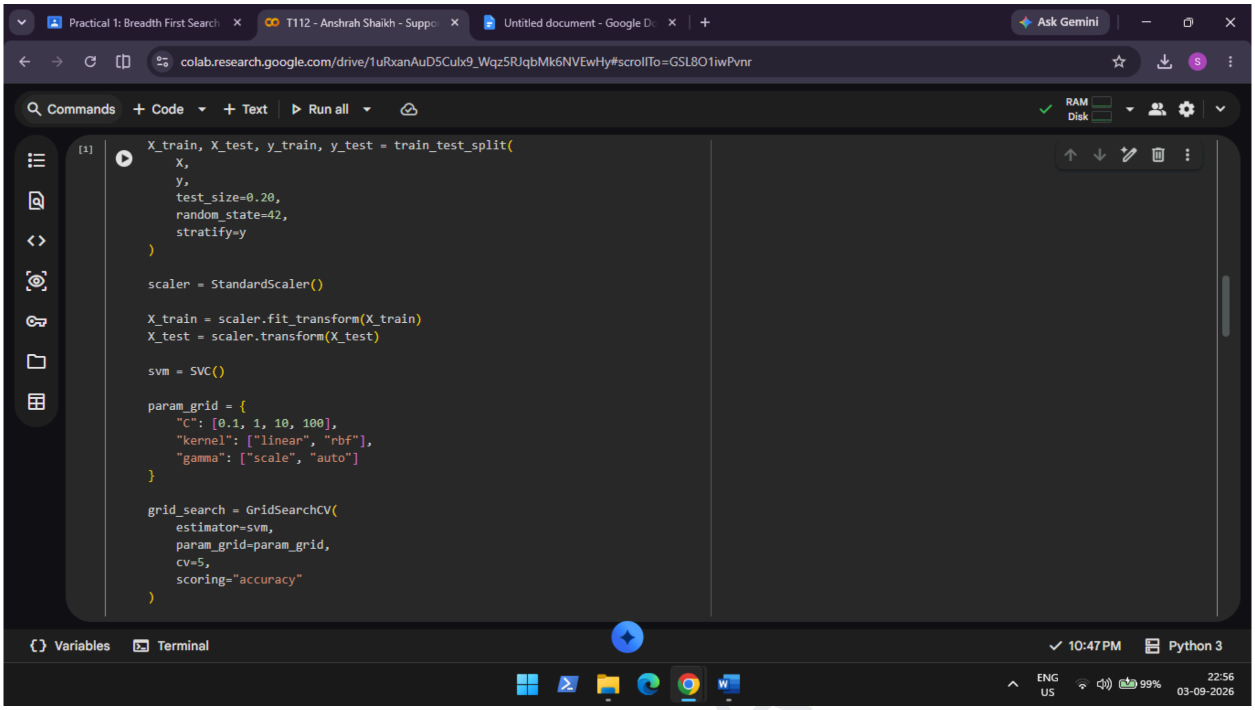


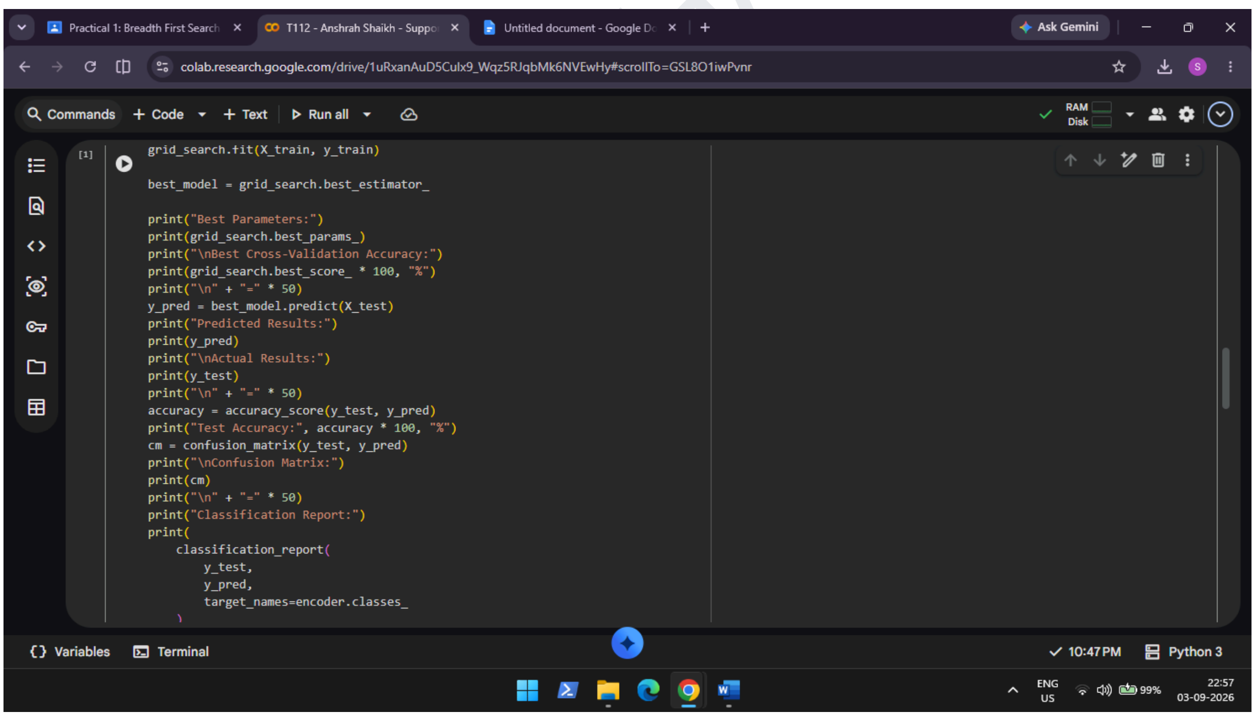


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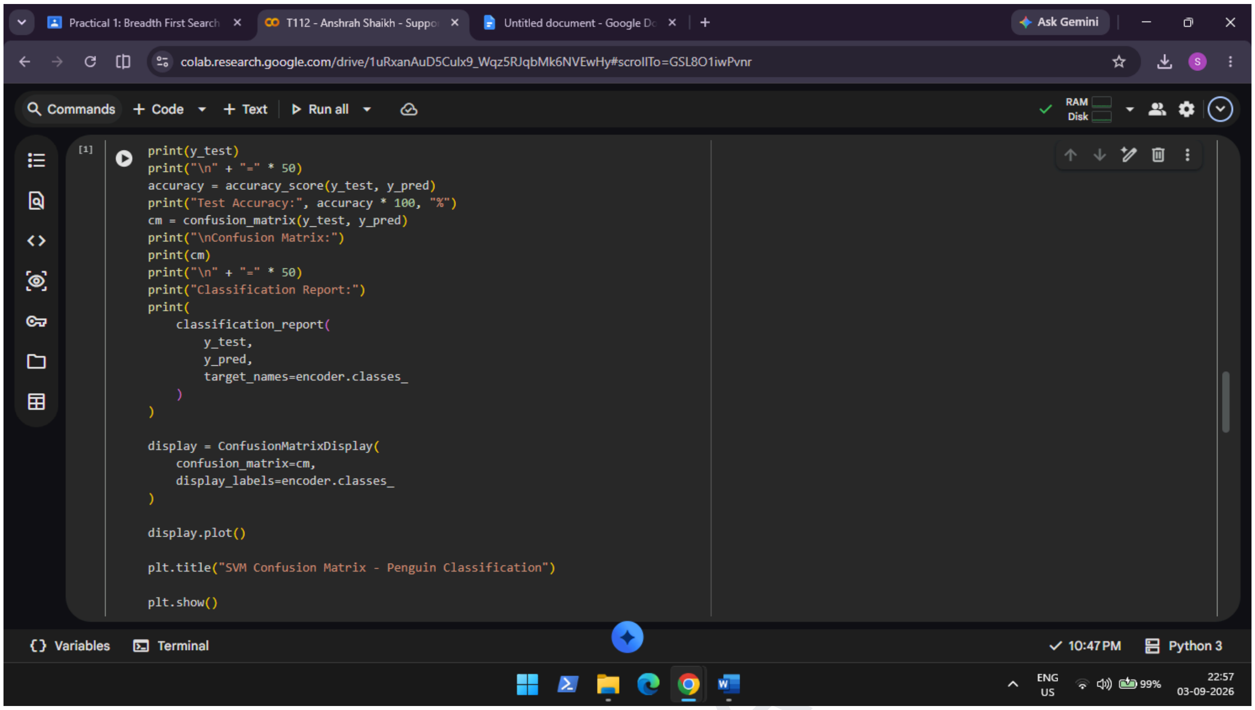
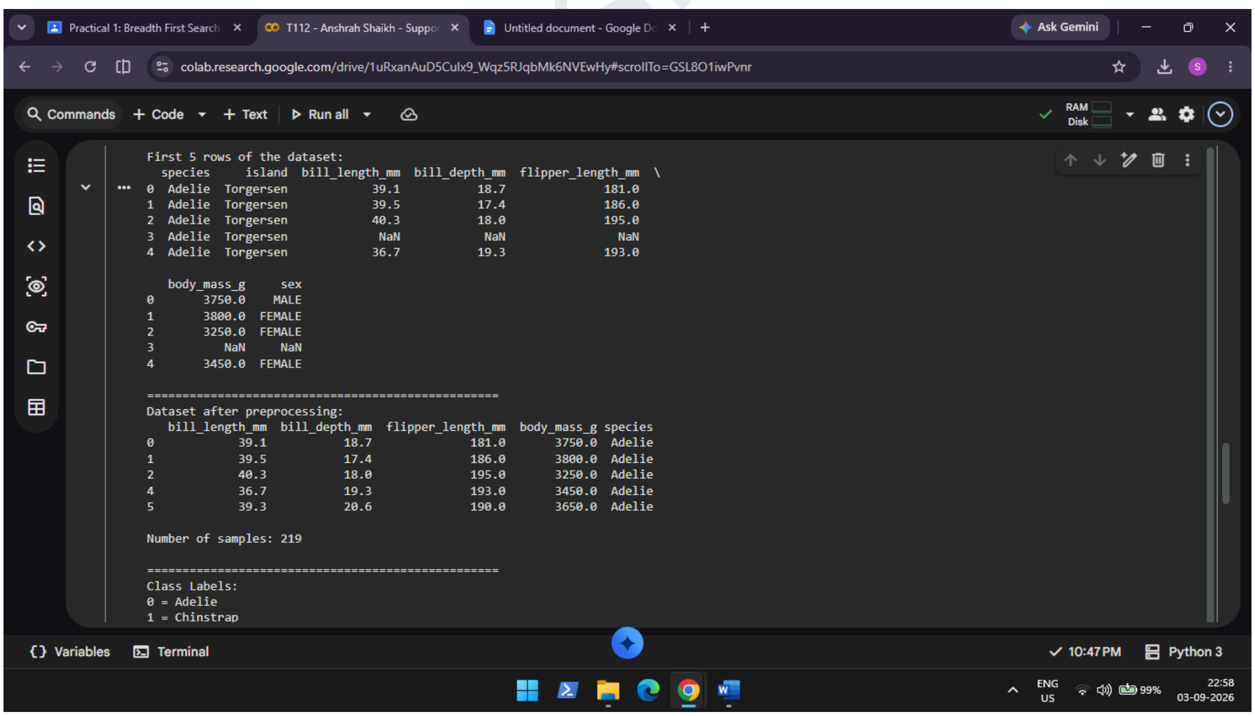


```python
print(y_test)
print("\n" + "=" * 50)
accuracy = accuracy_score(y_test, y_pred)
print("Test Accuracy:", accuracy * 100, "%")
cm = confusion_matrix(y_test, y_pred)
print("\nConfusion Matrix:")
print(cm)
print("\n" + "=" * 50)
print("Classification Report:")
print(
    classification_report(
        y_test,
        y_pred,
        target_names=encoder.classes_
    )
)

display = ConfusionMatrixDisplay(
    confusion_matrix=cm,
    display_labels=encoder.classes_
)

display.plot()

plt.title("SVM Confusion Matrix - Penguin Classification")

plt.show()
```

OUTPUT:



```
First 5 rows of the dataset:
  species     island  bill_length_mm  bill_depth_mm  flipper_length_mm  \
0  Adelie  Torgersen            39.1           18.7              181.0
1  Adelie  Torgersen            39.5           17.4              186.0
2  Adelie  Torgersen            40.3           18.0              195.0
3  Adelie  Torgersen             NaN            NaN                NaN
4  Adelie  Torgersen            36.7           19.3              193.0

   body_mass_g     sex
0       3750.0    MALE
1       3800.0  FEMALE
2       3250.0  FEMALE
3          NaN     NaN
4       3450.0  FEMALE

==================================================
Dataset after preprocessing:
   bill_length_mm  bill_depth_mm  flipper_length_mm  body_mass_g species
0            39.1           18.7              181.0       3750.0  Adelie
1            39.5           17.4              186.0       3800.0  Adelie
2            40.3           18.0              195.0       3250.0  Adelie
4            36.7           19.3              193.0       3450.0  Adelie
5            39.3           20.6              190.0       3650.0  Adelie

Number of samples: 219

==================================================
Class Labels:
0 = Adelie
1 = Chinstrap
```

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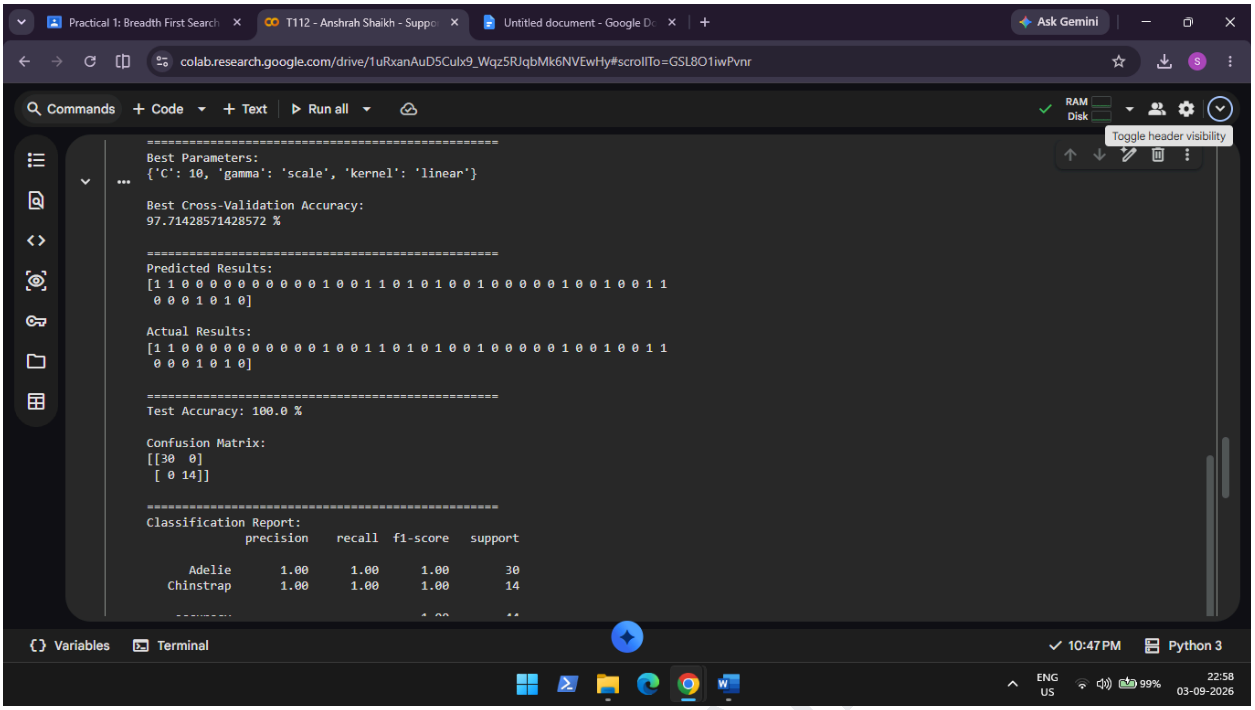


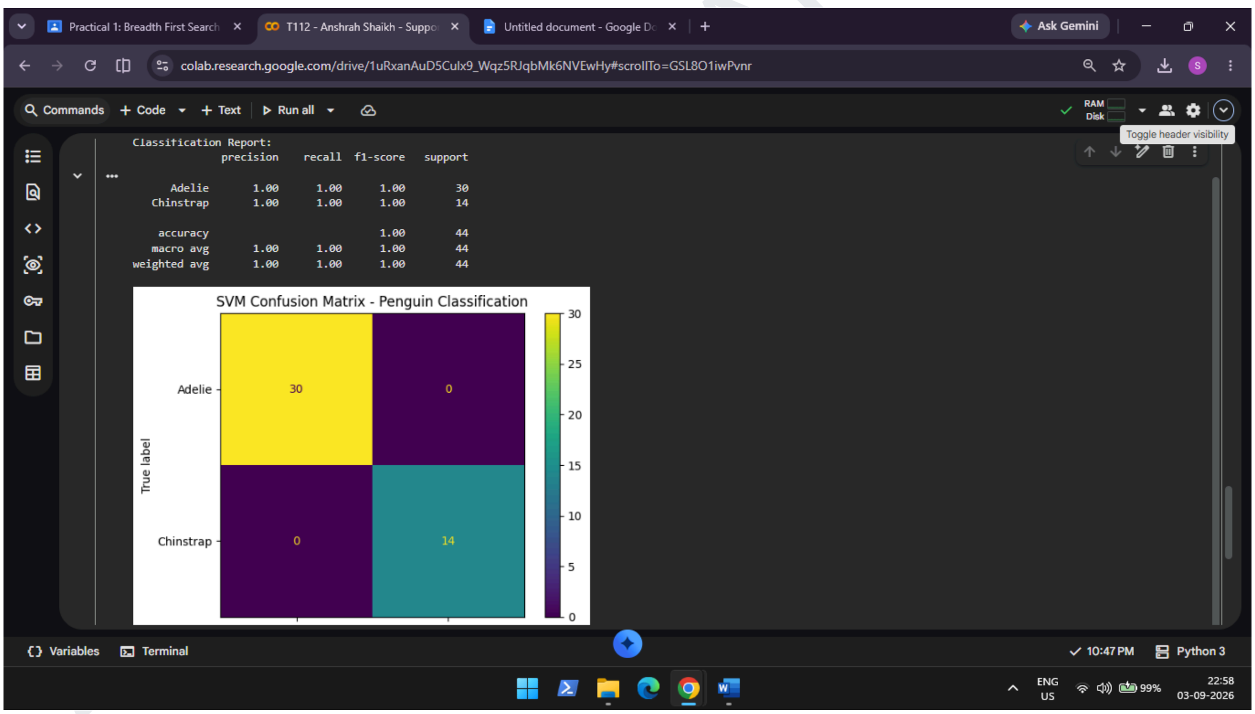


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